

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** to guide strategic business decisions





Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating only

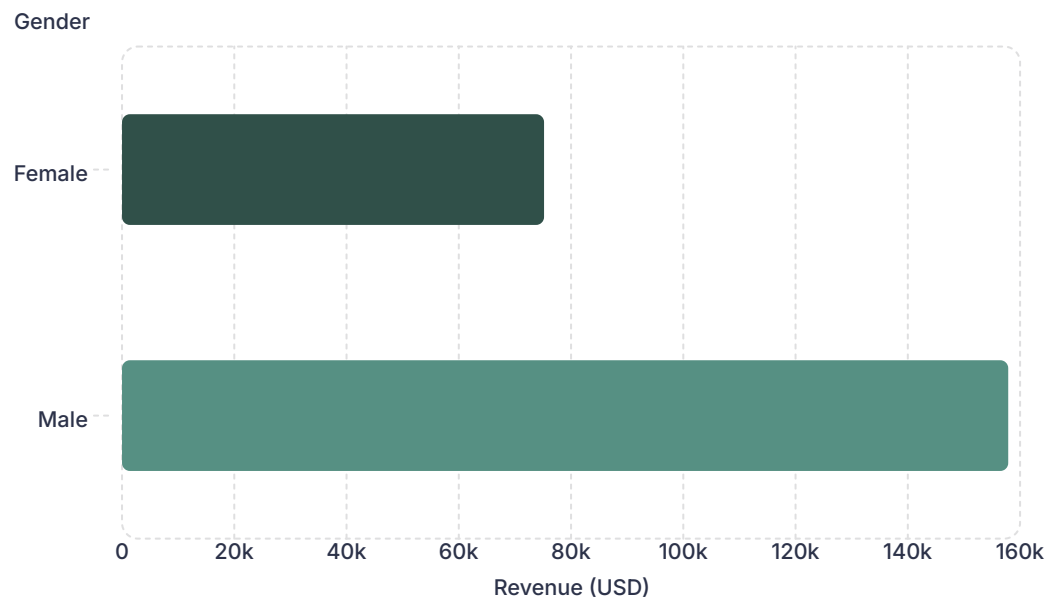
Key features: demographics, purchase details, shopping behavior, subscription status

Data Preparation in Python



- Clean & Standardize
Snake_case columns, median imputation for Review Rating
- Feature Engineering
Created `age_group` and `purchase_frequency_days`
- Database Load
Cleaned DataFrame loaded into PostgreSQL for SQL analysis

Revenue by Gender



Key Finding

Male customers generate **2.1x more revenue** than female customers.

Male: \$157,890 · Female: \$75,191



Top Products & Shipping Insights

★ Top-Rated Products

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)

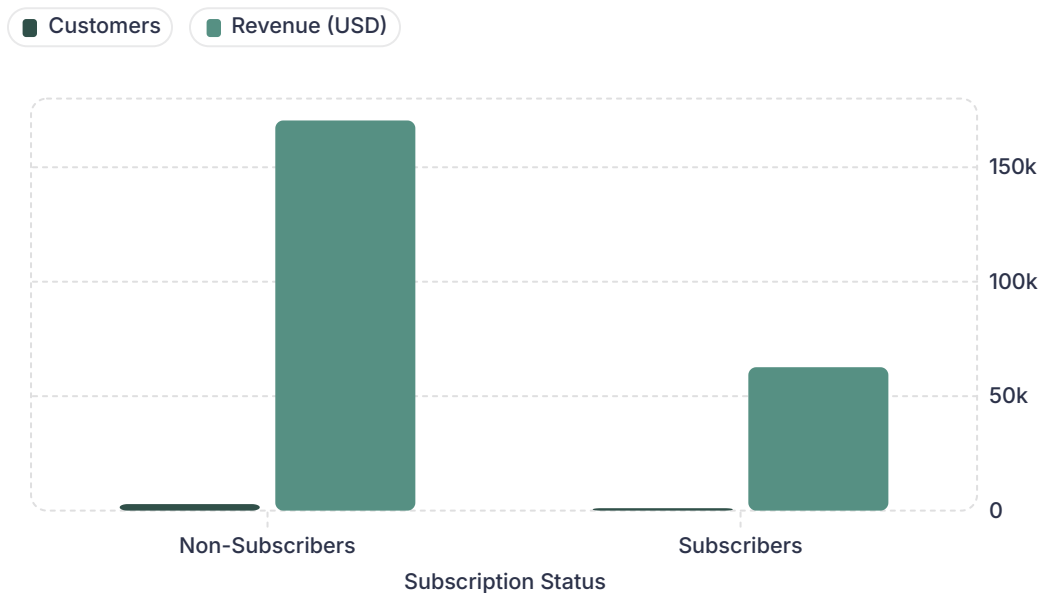
🚚 Shipping Type

Express avg: **\$60.48** vs Standard avg: **\$58.46**

💰 Discount Users

839 customers used discounts and spent above average

Subscription Behavior

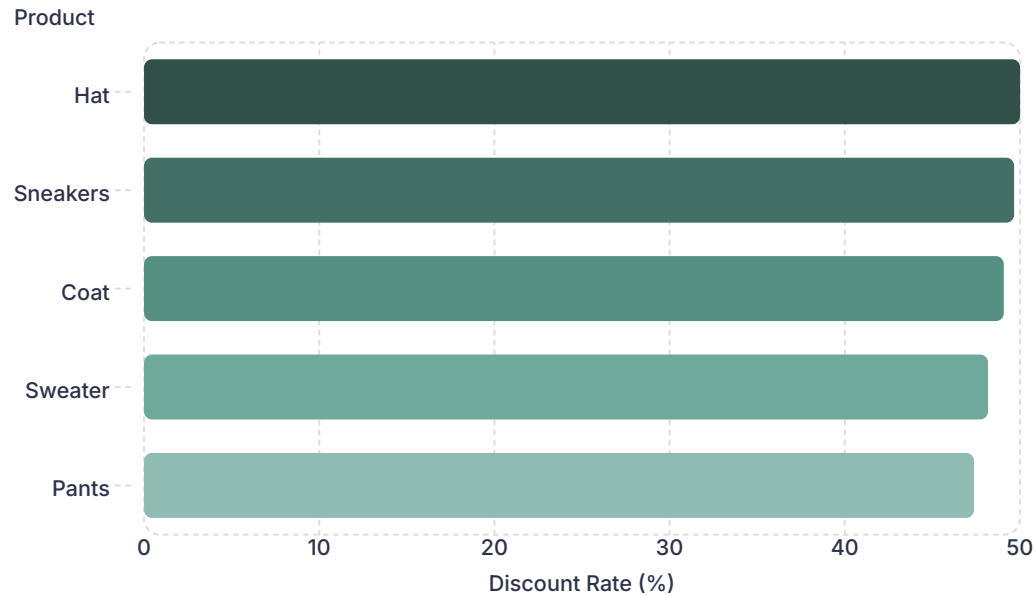


Subscribers vs Non-Subscribers

Non-subscribers drive **73%** of total revenue despite similar average spend (~\$59).

i Only 27% of customers are subscribed — significant growth opportunity.

Discount-Dependent Products



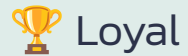
Key Insight

Top 5 products all have **~47–50% discount rates** — heavy reliance on promotions.



Review discount policy to balance sales lift with margin control.

Customer Segmentation



Loyal

3,116 customers



Returning

701 customers



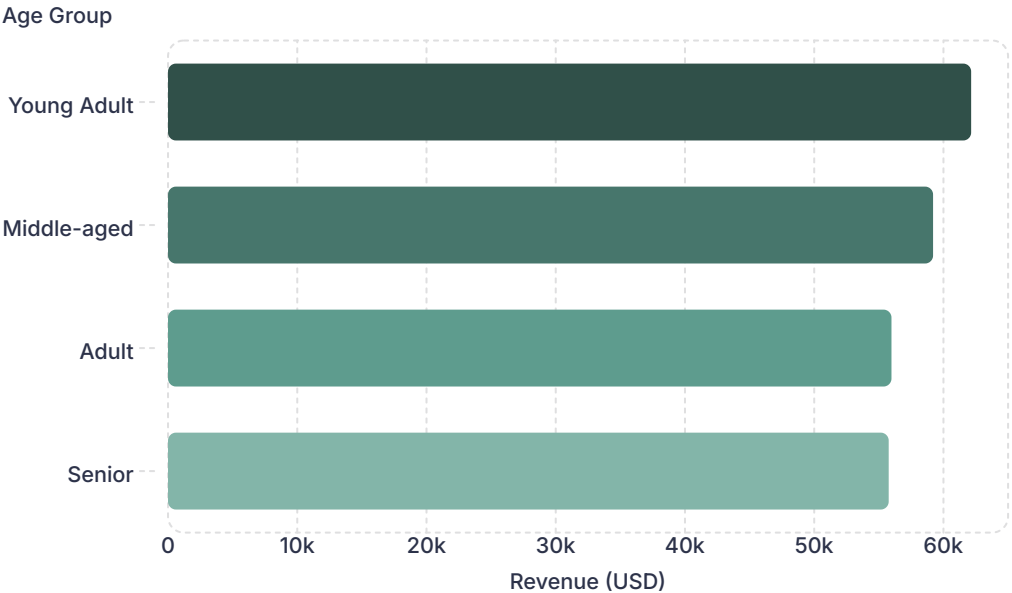
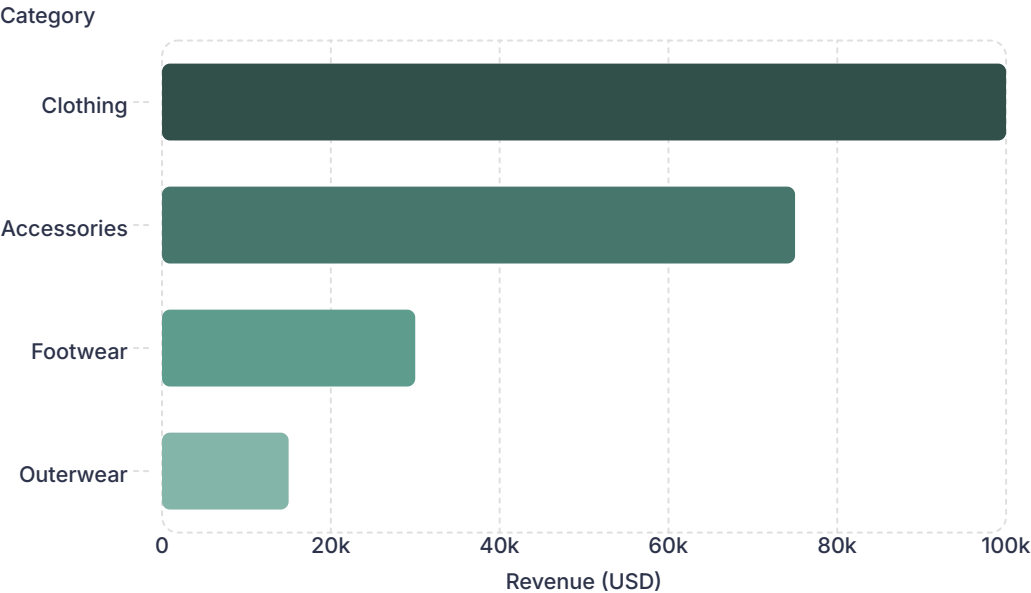
New

83 customers

958 repeat buyers (>5 purchases) are subscribed; 2,518 are not — a key conversion target.



Revenue by Category & Age Group



Clothing dominates revenue. Young Adults lead age-group spending.



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users